Higher homotopy groups

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Introduction

From the first course of topology we have come to awareness about the fundamental group, known as π_1. In these pages we are going to define the superior homotopy group π_n, a kind of generalization of π_1, in order to prove the *Whitehead's* theorem. Lastly we will compute $\pi_n(S^k)$ for some n and k.
In the following pages we will use the following notation: for any $X_1 \subset \cdots \subset X_n$ and $Y_1 \subset \cdots \subset Y_n$ topological spaces will indicate with $f : (X_n, \ldots, X_1) \longrightarrow (Y_n, \ldots, Y_1)$ a function such that $f(X_i) \subset Y_i \ \forall i \in \{1, \ldots, n\}$. Moreover, an homotopy $H(x, t)$ between $f, g : (X_n, \ldots, X_1) \longrightarrow (Y_n, \ldots, Y_1)$ is an homotopy between f, g such that $H(X_i, t) = Y_i$ for all $t \in I$ and $i \in \{1, \ldots, n\}$, and we will write $f \simeq g$. We can now define $[(X_n, \ldots, X_1), (Y_n, \ldots, Y_1)] := \{f \text{ function such that } f : (X_n, \ldots, X_1) \longrightarrow (Y_n, \ldots, Y_1)\}/\simeq$.
We will write $I^n := [0, 1]^n := \prod_{i=1}^{n} [0, 1] \subset \mathbb{R}^n$.

1 First definitions

Definition 1.1. The $n-$th homotopy set of X topology space with base point $x_0 \in X$ is $\pi_n(X, x_0) := \{\overline{f} \mid f : I^n \longrightarrow X,\ f(\partial I^n) = x_0\}$, where the classes are defined up to omothopy.

Proposition 1.2. For $n \geq 1$ $\pi_n(X, x_0)$ is a group with the following product:

$$\overline{f}, \overline{g} \in \pi_n(X, x_0) \implies (f * g)(t_1, \ldots, t_n) := \begin{cases} g(2t_1, t_2, \ldots, t_n), 0 \leq t_1 \leq 1/2 \\ f(2t_1, t_2, \ldots, t_n), 1/2 \leq t_1 \leq 1 \end{cases}$$

One may object that the definition of the group structure prefers the coordinjate t_1 over the other one. We could also define a product as follows:

$$(f *_i g)(t_1, \ldots, t_n) := \begin{cases} g(t_1, \ldots, 2t_i, \ldots, t_n), 0 \leq t_i \leq 1/2 \\ f(t_1, \ldots, 2t_i, \ldots, t_n), 1/2 \leq t_i \leq 1 \end{cases}$$

The following results will demonstrate that all the products upon induce the same product on the group π_n.

Lemma 1.3. For any $a, b, c, d : (I^n, \partial I^n) \longrightarrow (X, x_0)$ maps we have

$$(a *_i b) * (c *_i d) = (a * c) *_i (b * d)$$

Proof. Let be a, b, c, d as above, then:

$$(a *_i b) * (c *_i d) = \begin{cases} a *_i b \text{ if } t_1 \in [0, 1/2] = \begin{cases} a \text{ if } t_1 \in [0, 1/2] \text{ and if } t_i \in [0, 1/2] \\ b \text{ if } t_1 \in [0, 1/2] \text{ and if } t_i \in [1/2, 1] \end{cases} \\ c *_i d \text{ if } t_1 \in [1/2, 1] = \begin{cases} c \text{ if } t_1 \in [1/2, 1] \text{ and if } t_i \in [0, 1/2] \\ d \text{ if } t_1 \in [1/2, 1] \text{ and if } t_i \in [1/2, 1] \end{cases} \end{cases} =$$

$$= \begin{cases} a * c \text{ if } t_i \in [0, 1/2] \\ b * d \text{ if } t_i \in [1/2, 1] \end{cases} = (a * c) *_i (b * d)$$

□

The following proposition is also known as "*Eckman-Hilton trick*":

Proposition 1.4. Let S be a set with two operation $\#, + : S \times S \longrightarrow S$ having a common unit e. Suppose also that for any $a, b, c, d \in S$

$$(a + b)\#(c + d) = (a\#c) + (b\#d)$$

Then $\#$ and $+$ define the same commutative and associative operation on S.

Proof. By taking $b = c = e$ we have that for all $a, d \in S$ $a + d = a\#d$. By taking $a = d = e$ we have that for all $b, c \in S$ $b\#c = c+d$. By taking $b = e$ we have associativity property. □

Corollary 1.5. The groups $\pi_n(X, x_0)$ are abelians for $n \geq 2$ and the products $*$ and $*_i$ define the same operation on the groups for all $n \in \mathbb{N}$ and for all $i \in \{1, \ldots, n\}$.

Theorem 1.6. For any X, Y topological space, we have the following results, as for the fundamental group:

1. $X \simeq Y$, then $\pi_k(X) \cong \pi_k(Y)$;
2. for all $f : X \longrightarrow Y$ we have an induced function $f_{*,n} : \pi_n(X) \longrightarrow \pi_n(Y)$;
3. for any $f : X \longrightarrow Y$ and $g : Y \longrightarrow Z$ we have $(g \circ f)_{*,n} = g_{*,n} \circ f_{*,n}$

The covering spaces gain a more central role in the study of the homotopy groups for higher dimension.

Remark 1.7. From the omeomorphism $I^n/\partial I^n \longrightarrow S^n$ we can state the existence of an isomorphism between $[(S^n, s_0), (X, x_0)] \cong \pi_n(X, x_0)$.

Theorem 1.8. A covering space projection $p : (\overline{X}, \overline{x}_0) \longrightarrow (X, x_0)$ induces isomorphism $p_* : \pi_n(\overline{X}, \overline{x}_0) \longrightarrow \pi_n(X, x_0)$ for all $n \geq 2$.

Proof. Let be $f \in \overline{f} \in [(S^n, s_0), (X, x_0)]$, by proposition 1.33 of *Hatcher* and because S^n is simply connetted for $n \geq 2$ we have a lifting map $f_1 : (S^n, s_0) \longrightarrow (\overline{X}, \overline{x}_0)$, hence we have $f = p \circ f_1$ and so we have $p_*(f_1) = f$. So we have the surjectivity.
Let's prove the injectivity: let be f such that $[f] \in ker(p_*)$, so we have that $[p \circ f] = 0$ and so $p \circ f \simeq c_{x_0}$. Hence exists an homotopy $H : S^n \times [0, 1] \longrightarrow X$ such that

$$H(x, 0) = p \circ f \text{ and } H(x, 1) = c_{x_0}$$

By proposition 1.30 from *Hallen Hatcher* (the homotopy lifting property) we have that exists an unique homotopy $\overline{H} : S^n \times [0, 1] \longrightarrow \overline{X}$ such that $p \circ \overline{H} = H$. By construction we have that:

- $p \circ \overline{H}(x, 0) = H(x, 0) = p \circ f$ and by uniqueness of $\overline{H}$ we have that $\overline{H}(x, 0) = f$;
- $p \circ \overline{H}(x, 1) = H(x, 1) = c_{x_0} = p \circ c_{\overline{x_0}}$ where $c_{\overline{x_0}} : S^n \longrightarrow \overline{X}$ is the costant map on $\overline{x_0}$. By uniqueness of $\overline{H}$ we have that $\overline{H}(x, 1) = c_{\overline{x_0}}$.

So, exists an homotopy between $c_{\overline{x_0}}$ and f, and so $0 = [f] \in \pi_n(\overline{X}, \overline{x_0})$. $\square$

Corollary 1.9. $\pi_n(X, x_0) = \{0\}$ with $n \geq 2$ whenever X admit a contractible universal cover. In particular $\pi_n(S^1) = \{0\}$ for all $n \geq 2$.

2 Long exact sequence

Definition 2.1. For X topological space and $A \subset X$ susbset and $x_0 \in X$ element we can define $\pi_n(X, A, x_0) := [(I^n, \partial I^n, J^{n-1}), (X, A, x_0)]$ where $J^n := I^n \times \{0\} \cup \partial I^n \times I$ for $n \geq 1$ and we define $\partial I^1 := \{0, 1\}$ and $J^0 := \{1\}$.

Remark 2.2. If $n \geq 2$ $\pi_n(X, A, x_0)$ is a group with the operation $*_i$ for all $i \leq n$ and is abelian for $n \geq 3$ (it can be shown as before with the "Eckman-Hilton trick"). Instead for $n = 1$ we have $I = [0, 1]$ and so $\pi_1(X, A, x_0)$ is the set of the classes of path from a random point in A to a fixed point x_0, so if $|A| > 1$ the operations are not well defined.

As in the homology case we have a long exact sequence:

Theorem 2.3. Let (X, A, x_0) a triple as above, exists homomorfisms $\partial : \pi_n(X, A) \longrightarrow \pi_{n-1}(A, x_0)$ such that the following sequence is exact:

$$\cdots \to \pi_{n+1}(X, A) \xrightarrow{\partial} \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A) \xrightarrow{\partial} \cdots \xrightarrow{\partial} \pi_0(A, x_0) \xrightarrow{i_*} \pi_0(X, x_0)$$

In the following pages we will write $\pi_n(X, A)$ instead of $\pi_n(X, A, x_0)$.
In the above diagram not all the maps are homomorphisms. In fact, $\pi_1(X, A)$, $\pi_0(A, x_0)$ and $\pi_0(X, x_0)$ are just set, not groups. However we can extend the notion of exactness by defining $ker(f) := \{x_0\}$ for avery map f with domain or codomain that is not a group.

Corollary 2.4. Given a $x_0 \in A \subset X$ a topological space such that $X \simeq *$, then there are isomorphisms $\pi_n(X, A) \cong \pi_{n-1}(A)$.

Example 2.5. We can apply the above corollary to the case of the reduced cone CA. We also have the inclusion $A \longrightarrow CA$, so we can apply the corollary and conclude that $\partial : \pi_{n+1}(CA, A) \longrightarrow \pi_n(A, *)$ is an isomorphism.

3 Whitehead's theorem

Definition 3.1. A map $f : X \longrightarrow Y$ is a weak homotopy equivalence if the induced maps

$$f_* : \pi_k(X, x_0) \longrightarrow \pi_k(Y, f(x_0))$$

are isomorphisms $\forall k \in \mathbb{N}$ and for all $x_0 \in X$ base points.

Remark 3.2. If one weakens this condition by considering a single base point only, then one obtain a different notion.

In order to establish the $Whitehead$'s theorem we are going to state two lemmas and one definition:

Definition 3.3. For X and Y CW-complexes a map $f : X \longrightarrow Y$ satisfying $f(X^n) \subset Y^n$ for all $n \in \mathbb{N}$ is called cellular map.

Theorem 3.4. (Cellular approximation theorem) Every map $f : X \longrightarrow Y$ of CW-complexes is homotopic to a cellular map.

Lemma 3.5. If X, Y are CW-complexes and $f : X \longrightarrow Y$ is homotopic to a cellular map then (M_f, X) is a CW pair.

Lemma 3.6. Let (X, A) be a CW pair and let (Y, B) a pair of space with $B \neq \emptyset$ such that $\pi_n(Y, B) = 0$ for all $n \in \mathbb{N}$. Then any map $f : (X, A) \longrightarrow (Y, B)$ is homotopic relative A to a map g such that $g(X) \subset B$ [1]. In particular $f(a) = g(a)$ $\forall a \in A$.

Theorem 3.7. (Whitehead's theorem) Let $f : X \longrightarrow Y$ be a weak equivalence between CW complexes. Then f is a homotopy equivalence.

Proof. By the mapping cylinder construction we have that the retraction $r : M_f \longrightarrow Y$ is an homotopic equivalence, so it induces the following isomorphism $r_{*,n} : \pi_n(M_f) \longrightarrow \pi_n(Y)$ for all n in the long exact sequence of homotopy group of the pair (M_f, X) (theorem 2.3).

$$\cdots \to \pi_{n+1}(M_f, X) \xrightarrow{\partial_{n+1}} \pi_n(X) \xrightarrow{i_{*,n}} \pi_n(M_f) \xrightarrow{j_{*,n}} \pi_n(M_f, X) \to \cdots$$

Where $i : X \longrightarrow M_f$ is the inclusion of subcomplex such that $f = r \circ i$. Hence we have $f_{*,n} = r_{*,n} \circ i_{*,n}$, and we also have $(r_{*,n})^{-1} \circ f_{*,n} = i_{*,n}$ so also $i_{*,n}$ is an isomorphism.
Now we have that $\{0\} = ker(i_{*,n}) = Im(\partial_{n+1})$, so $\partial_{n+1} = 0$ is the null map for $n \geq 0$, but $\pi_n(M_f) = Im(i_{*,n}) = ker(j_{*,n})$ and so also $j_{*,n} = 0$ is the null map. Since $Im(j_{*,n}) = ker(\partial_n) = \pi_n(M_f, X)$ we have that $j_{*,n}$ is surjective. So we have that $j_{*,n}$ is surjective and null, so $\pi_n(M_f, X) = 0$ for all $n \geq 1$.
By theorem 3.4 and lemma 3.5 we have that (M_f, X) is a CW pair. Hence we can apply lemma 3.6 to the identity $id_{M_f} : (M_f, X) \longrightarrow (M_f, X)$. We have that $id_{M_f} \simeq g$ with $g : M_f \longrightarrow M_f$ such that $g|_X = id_X$ and $g(M_f) \subset X$, so we can write g as $g = i \circ h$ with $h : M_f \longrightarrow X$ such that $h|_X = id_X$. Hence $r \circ i = h|_X = id_X$. Hence i is an homotopy equivalence. Since we have that the composition of homotopy equivalences is an homotopy equivalence, then $f = r \circ i$ is an homotopy equivalence.

∎

[1] Exists H homotopy between f and g such that $H(a, t) = H(a, 0)$ $\forall a \in A$ and $\forall t \in I$.

Remark 3.8. The upper theorem does not states that two CW complexes with isomorphic groups are homotopy. We must have a function that induces isomorphisms between homotopic groups.
For example Let be $A = \mathbb{RP}^2$ and $B = S^2 \times \mathbb{RP}^\infty$. Their universal covers are S^2 and $S^2 \times S^\infty$, and we have that $\pi_n(S^2) \cong \pi_n(S^2 \times S^\infty)$ because S^∞ is contractible. By 1.8 we have that $\pi_n(A) = \pi_n(B)$ for $n \geq 2$. We also have that $\pi_1(A) = \mathbb{Z}/2\mathbb{Z} = \pi_1(B)$, so for all n we have that $\pi_n(A) = \pi_n(B)$. But $H_3(A) = \{0\}$, because there is not structure in higher dimension, and $H_3(B) \cong \mathbb{Z}/2\mathbb{Z}$. So they cannot be homotopic.

4 Homotopy groups for the k-sphere

Corollary 4.1. $\pi_n(S^k) = 0$ for $n < k$.

Proof. By remark 1.7 we can consider every map h such that $[h] \in \pi_n(S^k)$ as map $h : S^n \longrightarrow S^k$. We can build S^n and S^k by the CW structure with 1 0-cell and respectively 1 n-cell and 1 k-cell. Let f a map such that $[f] \in \pi_n(S^k)$, for theorem 3.4 exists a cellular map $g : S^n \longrightarrow S^k$ such that $f \simeq g$ and such that $g(X^r) \subset (Y^r)$ where X^r is the r-skeleton of S^n and Y^r is the r-skeleton of S^k. Since $X^n = S^n$ and for all $n < k$ $Y^n = Y^0 = \{pt.\}$, we have that $g(S^n) = g(X^n) \subset Y^n = \{pt.\}$. Hence g is costant, and so $f \simeq g = c_{\{pt.\}}$. So $[f] = 0$ and we have the thesis by the aribtrarity of f. $\square$

Definition 4.2. Let X be a topological space and $x_0 \in X$, we will say that X is n-connected if $\pi_j(X, x_0) = \{0\}$ $\forall j \leq n$

Now we are going to state the excision theorem for homotopy groups:

Theorem 4.3. Let X be a CW-complex and A, B subcomplexes such that $C := A \cap B \neq \emptyset$ and is connected. If (A, C) is m-connected and (B, C) is n-connected, then the map $\pi_i(A, C) \longrightarrow \pi_i(X, B)$ induced by inclusion is an isomrophism for $i < m+n$ and a surjection for $i = m + n$.

Corollary 4.4. (Freudenthal suspension theorem) Let X a connected CW-complex $(n-1)$-connected, then the map $\pi_i(X) \longrightarrow \pi_{i+1}(SX)$ is an isomorphism for $i < 2n-1$ and a surjection for $i = 2n - 1$.

Proof. $SX = C_+X \cup C_-X$ and $C_+X \cap C_-X = X'$, where X' is a copy in SX of X, so we have that C is connected by connecteness of X. We can consider the following long exact sequence:

$$\cdots \to \pi_{i+1}(C_+X) \to \pi_{i+1}(C_+X, X) \to \pi_i(X) \to \cdots$$

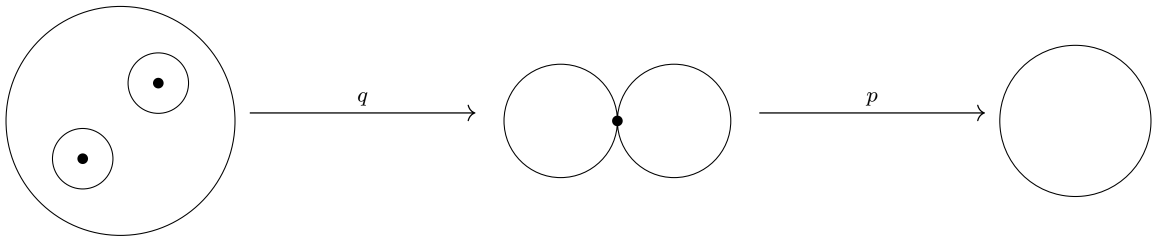
Since $C_\pm X$ are contractible we have that $\pi_i(C_\pm X) \cong \{0\}$ $\forall i \in \mathbb{N}$; moreover also $\pi_i(X) \cong \{0\}$ $\forall i \leq (n-1)$, because X is $(n-1)$-connected. By exacteness of the long sequence we have that $\pi_i(C_+X, X) = \{0\}$ $\forall i \leq (n-1)$: then (C_+X, X) is n-connected (analogously (C_-X, X)). Now we can use the theorem 4.3, so we have that $\pi_i(C_+X, X) \longrightarrow \pi_i(SX, C_-X)$ is an isomorphism for $i < 2n - 1$ and a surjection for $i = 2n - 1$.
Moreover, we can see the following sequence:

$$\cdots \to \pi_{i+1}(C_-X) \to \pi_{i+1}(SX) \to \pi_{i+1}(SX, C_-X) \to \pi_i(C_-X) \to \cdots$$

Since $\pi_i(C_-X) \cong \{0\}$ $\forall i \in \mathbb{N}$, we have $\pi_{i+1}(SX, C_-X) \cong \pi_{i+1}(SX)$. By example 2.5 we also have that $\pi_i(X) \cong \pi_{i+1}(C_+X, X)$.
Hence we have $\pi_i(X) \cong \pi_{i+1}(C_+X, X) \longrightarrow \pi_{i+1}(SX, C_-X) \cong \pi_{i+1}(SX)$ and then we also have that $\pi_i(X) \longrightarrow \pi_{i+1}(SX)$ is an isomorphism for $i < 2n - 1$ and a surjection for $i = 2n - 1$. $\square$

Lemma 4.5. $\forall n = 2^k$ with $k \in \mathbb{N}$ $\exists f : S^2 \longrightarrow S^2$ such that $deg(f) = n$

In the proof of the above lemma is used the local degree. We will not prove the lemma but we are going to give a sketch of the proof.
For any $f : S^2 \longrightarrow S^2$ such that for some point $y \in S^2$ $|f^{-1}(y)| < \infty$, we can define $deg\ f|x_i$ for $x_i \in f^{-1}(y)$ [2]. Let be B_1 and B_2 two disjoint open ball in S^2 and let $q : S^2 \longrightarrow S^2 \vee S^2$ be the quotient map that collaps the complement of the two balls to a point $p \notin B_1 \cup B_2$. Now, let $p : S^2 \vee S^2 \longrightarrow S^2$ identify all the copies to a single sphere.



If we define $f := pq$ we have that for almost all $y \in S^2$ we have that $f^{-1}(y) = \{x_1, x_2\}$ where $x_i \in B_i$. So the local degree $deg\ f|x_i = \pm 1$ because in B_i f is an homeomorphism. By composing a map g with f such that $deg(g) = -1$ we can assume that $deg\ f|x_i = 1$. And by the formula $deg(f) = deg\ f|x_1 + deg\ f|x_2$ we have that $deg(f) = 2$. Now we have that $deg(f^(k)) = (deg(f))^k = 2^k$ [3].
We can finally compute some non trivial homotopy group.

Corollary 4.6. $\pi_n(S^n) \cong \mathbb{Z}$

Proof. By corollary 4.1 we have that S^n is $(n-1)$-connected; moreover $S^k =$ [4]SS^{k-1}. Hence we can use the corollary 4.4:

$$\pi_1(S^1) \xrightarrow{g} \pi_2(S^2) \rightarrow \pi_3(S^3) \rightarrow \cdots$$

in this sequence the first map is surjective and the other ones are isomorphisms. Since $\pi_1(S^1) = \mathbb{Z}$, by the surjectivity of g we have that $\pi_n(S^n)$ are cyclic groups. We only have to prove that they have infinite cardinality. By lemma 4.5 we have that we can find infinite function with different degree, and we also know that the degree of a function is an homotopy invariant, so they have to stay in different homotopy classes. So the cardinality is infinity, so we have all isomorophisms.

□

[2]Formal definition on *page 136* of *Algebraic Topology*, *Hallen Hatcher*.
[3]We can also write down this for any $n \in \mathbb{N}$ (not only for power of 2), and also for every S^k (*page 136* of *Algebraic Topology*, *Hallen Hatcher*).
[4]The suspension of S^{k-1}.

Bibliography

1. *Algebraic Topology*, *Hallen Hatcher*;

2. Course material from lectures of the course *Algebraic Topology (Fall 2014)* in *Radboud University Nijmegen* holded by "*Ieke Moerdijk*" and "*Javier J. Gutiérrez*" `https://www.math.ru.nl/~gutierrez/algtop2014.htm#`.